



CONESTOGA

Connect Life and Learning

Title:

Olumide Akomolafe

9076433

Applied Computer Science & IT

Course Code: SECU8020

Course Instructor: Baljeet S. Bilkhu

Assignment Due Date: September 29

Assignment #1 – Kali and Metasploitable 2

The aim of this assignment is to explore how to identify vulnerabilities and exploit such vulnerabilities using Metasploit framework in Kali Linux to target a vulnerable Metasploitable virtual machine.

Task 1

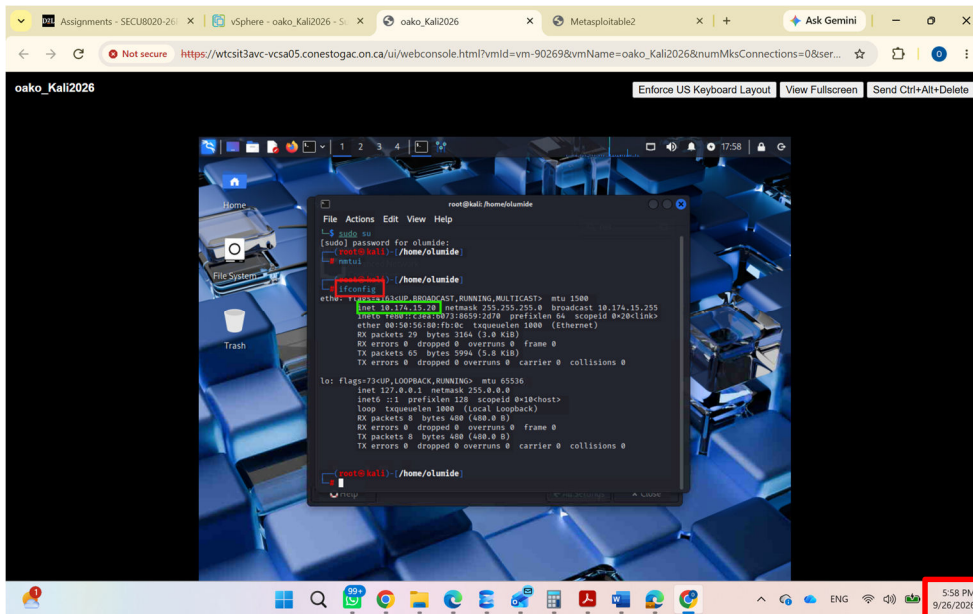


Image 1: Using the *ifconfig* to find the IP (10.174.15.20) address on Kali

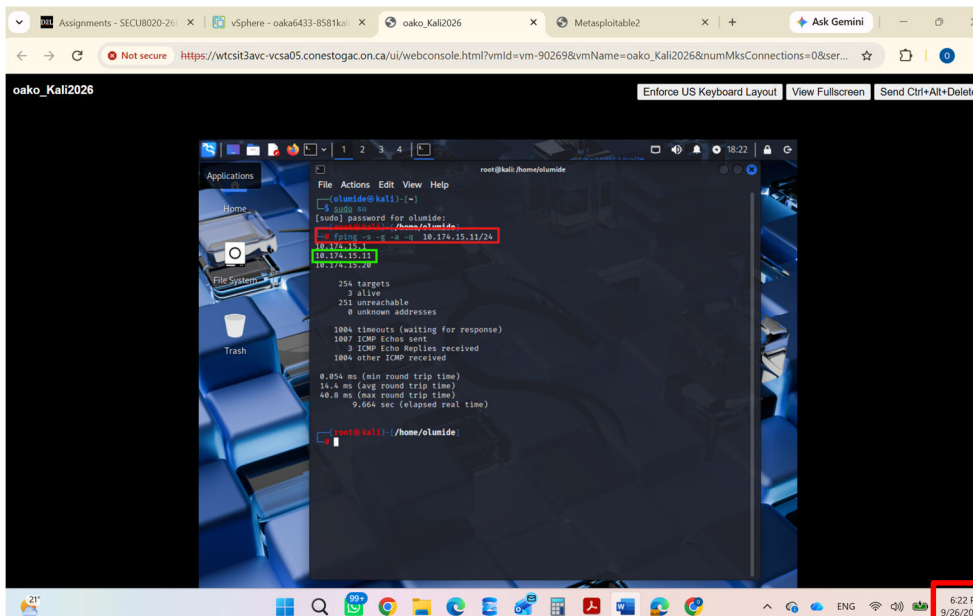


Image 2: Image 1: Using the *fping* to find the IP (10.174.15.11) address on Metasploitable2

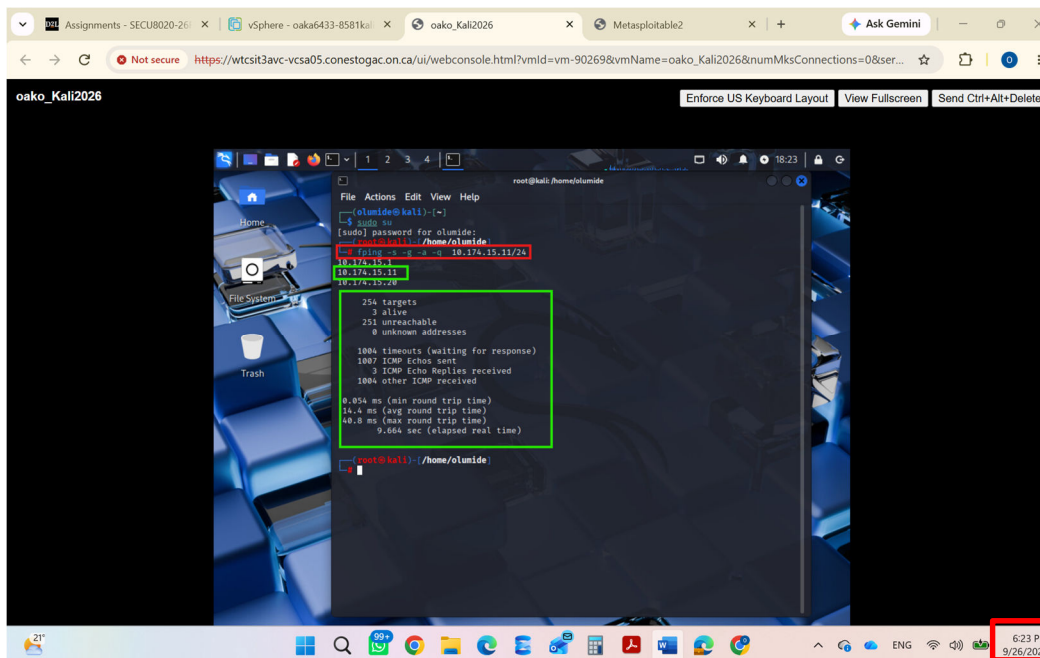


Image3: using `fping -s -g -a -q 10.174.15.11/24` to generate a target list, show alive targets, quite output and final stats

Task 2

Ping test

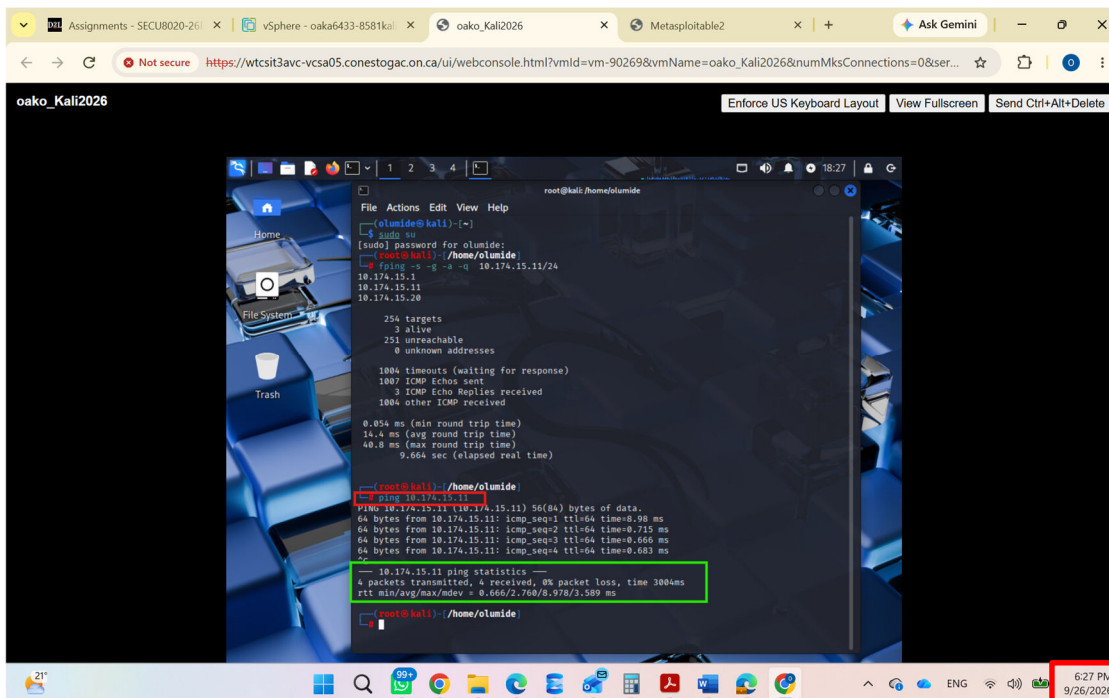


Image 4: ping 10.174.15.11 on Kali to test connection with Metasploitable 2 VM was successful

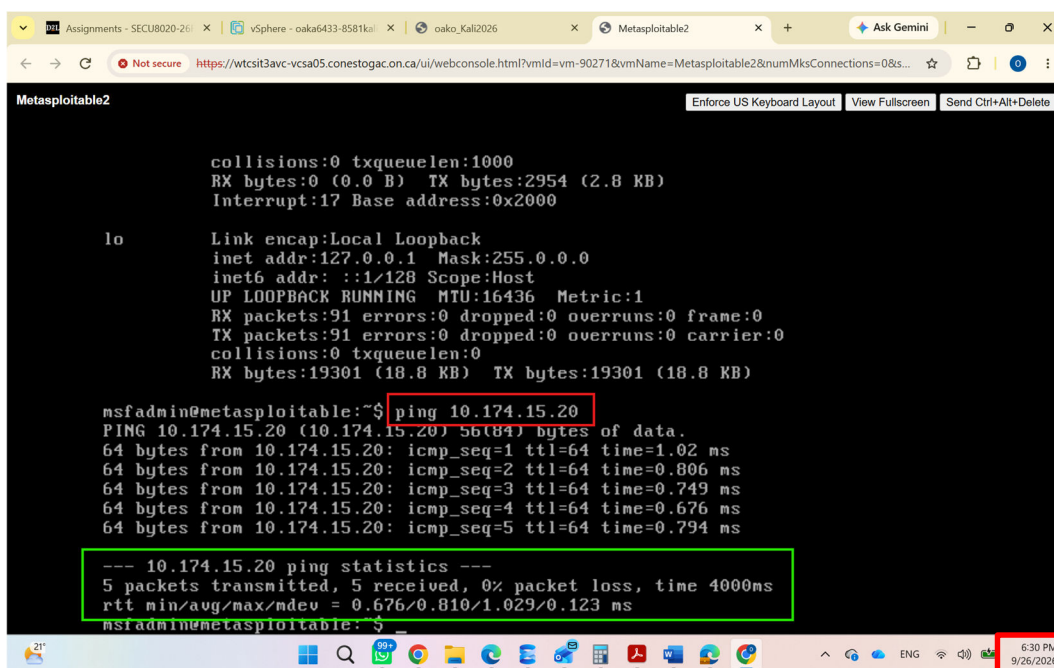


Image 5: ping 10.174.15.20 on Metasploitable to test connection with Kali VM was successful

Task 3

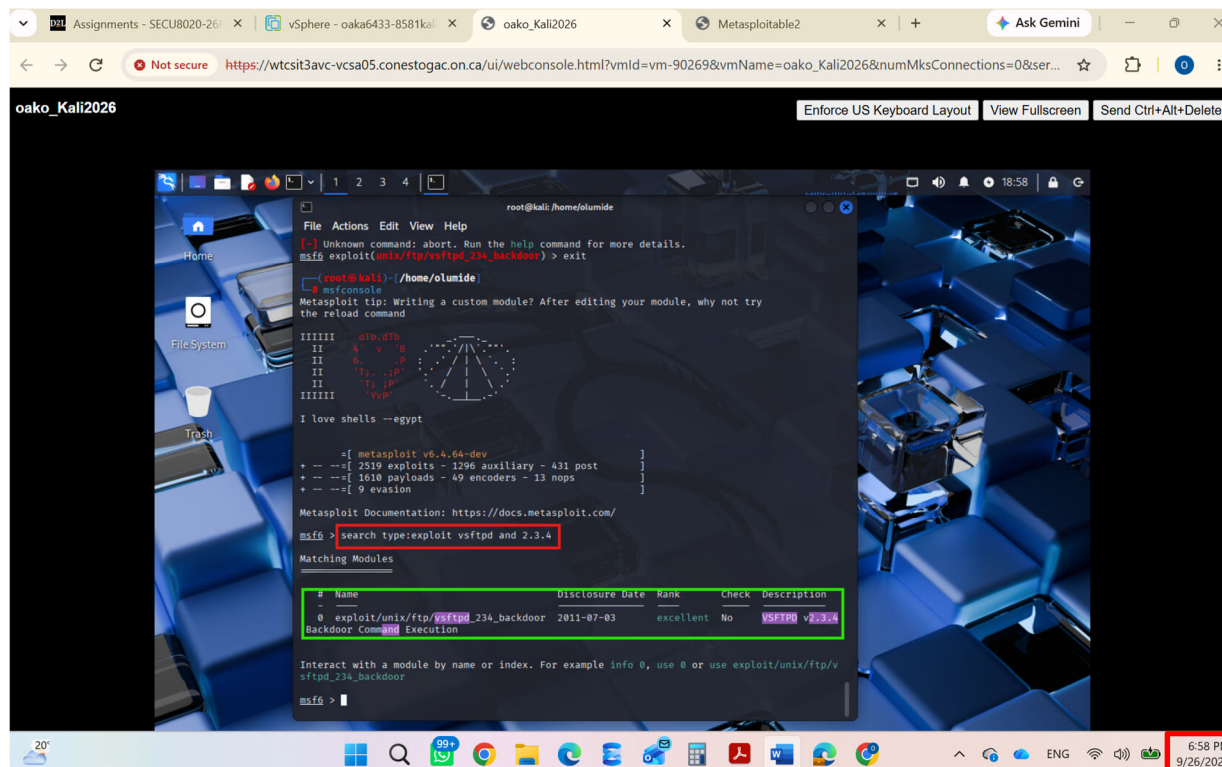


Image 6: using the “*search type:exploit vsftpd and 2.3.4*” command to search for an exploit in Metasploit framework.

Task 4

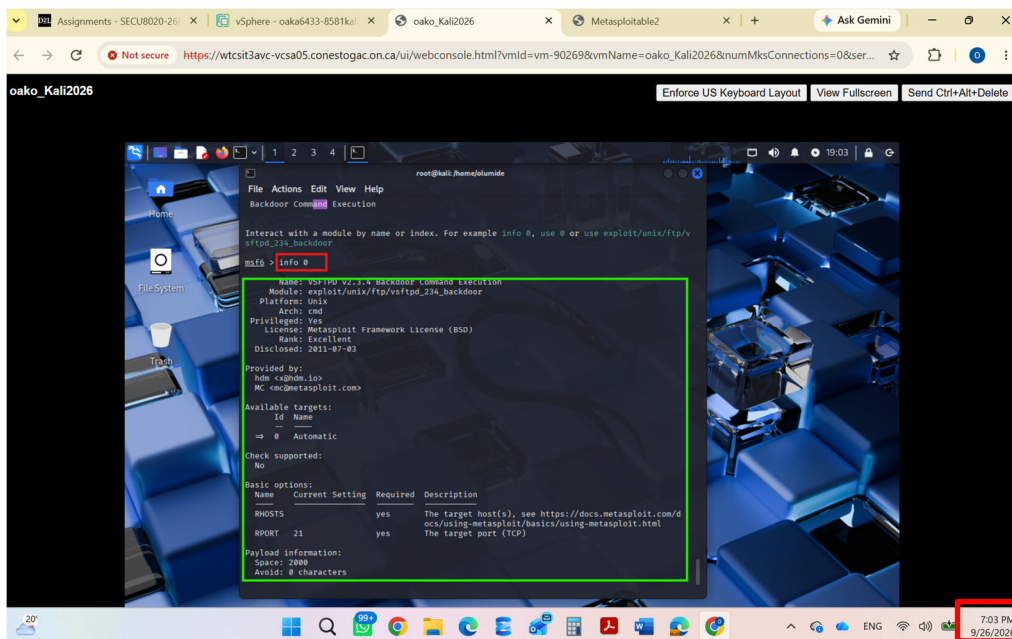


Image 7

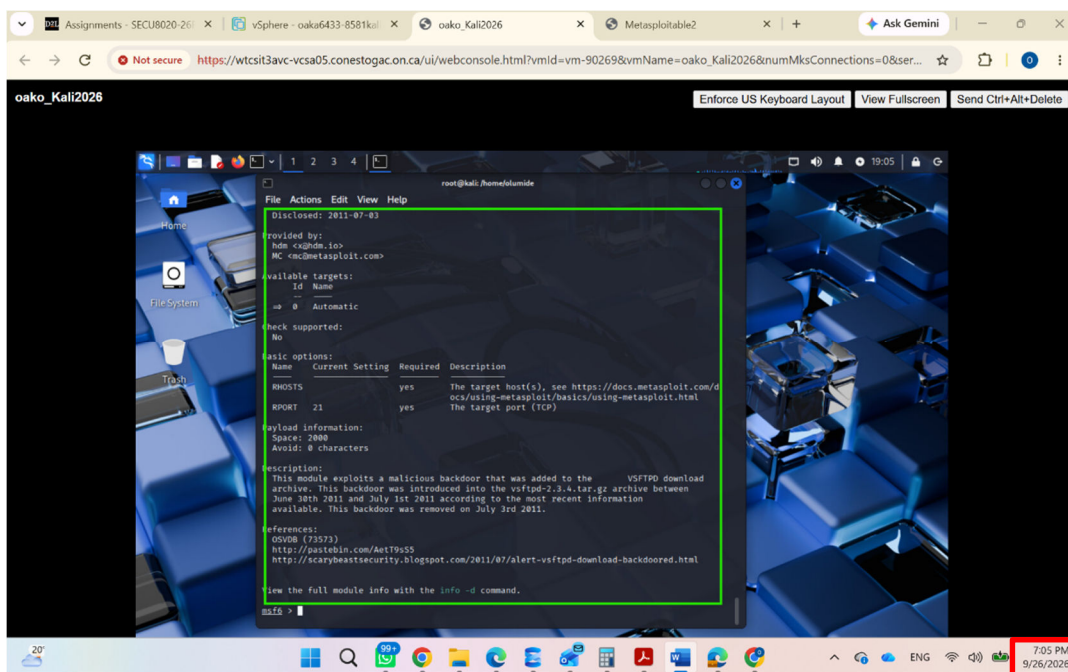


Image 8

Image 7 and 8 shows output of the *info 0* command in Metasploit framework to find out more information about the vsftpd_234 module.

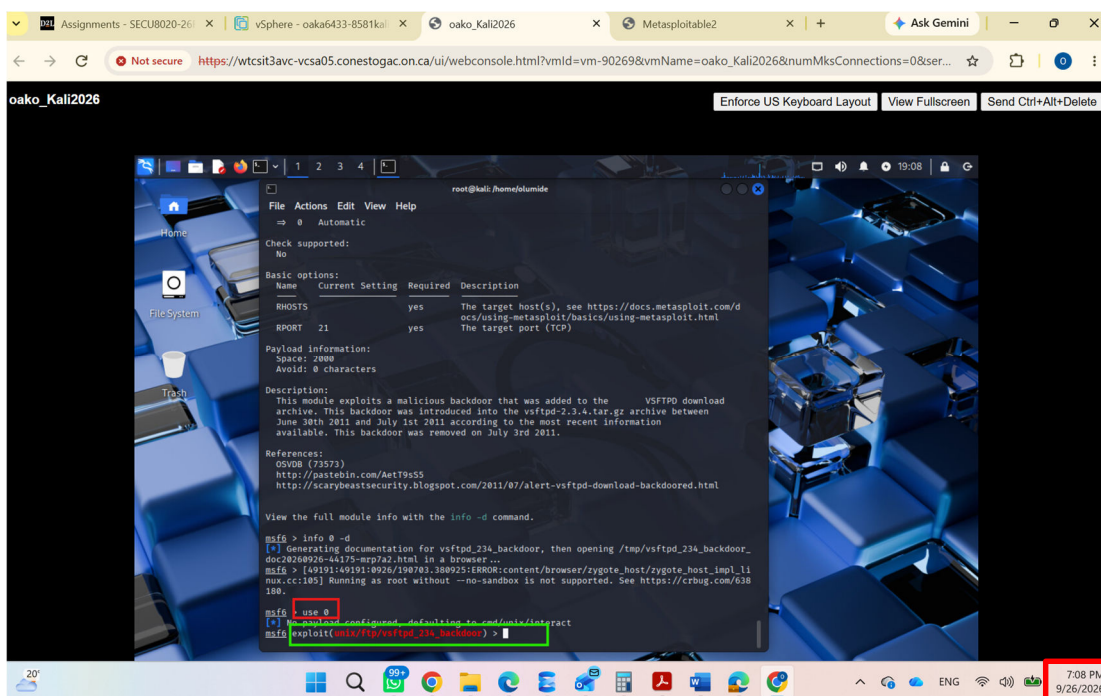


Image 9: vsftpd_234 module loaded from search into memory.

Task 5

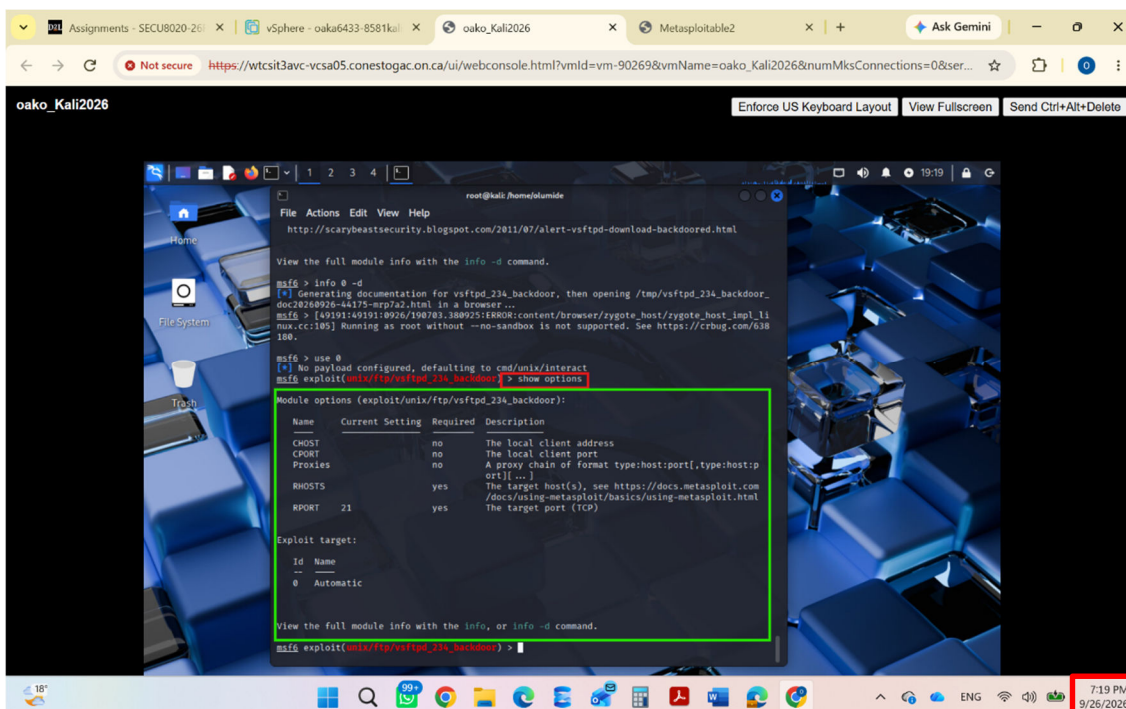


Image 10: show options command displays options associated with the loaded vsftpd_234 exploit from which we see that RHOST and RPORT can be used to take advantage of Metasploitable 2 VM.

Task 6

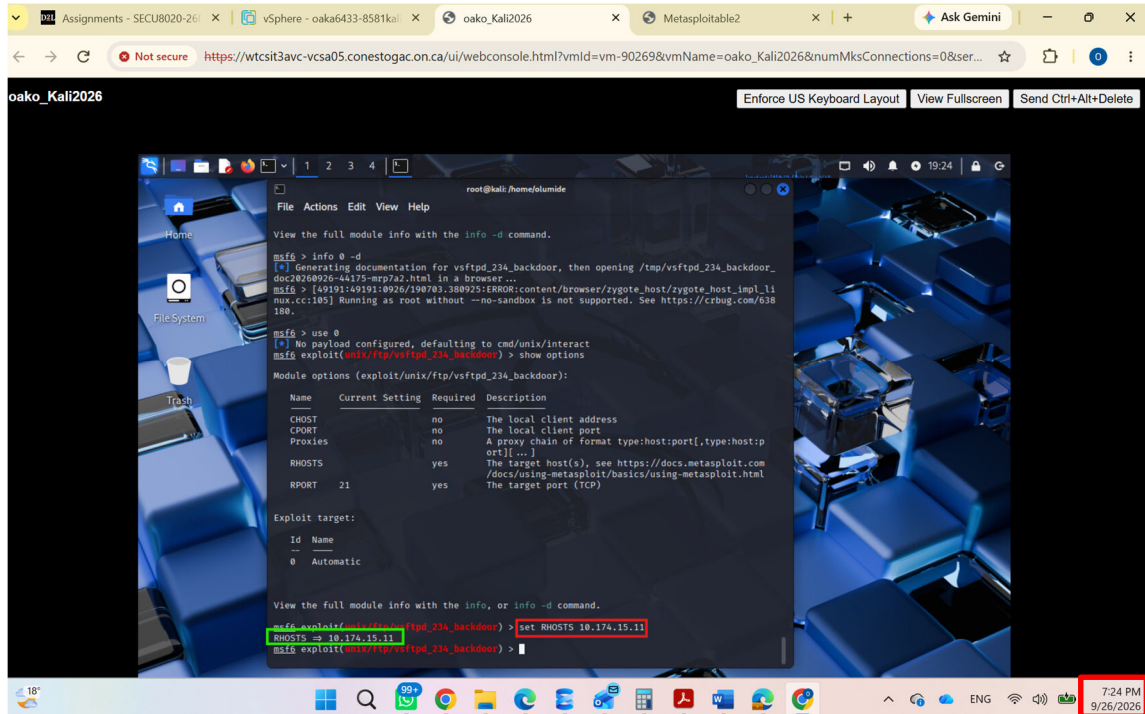


Image 11: set RHOSTS 10.174.15.11 command that will allow me exploit Metasploitable 2 VM.

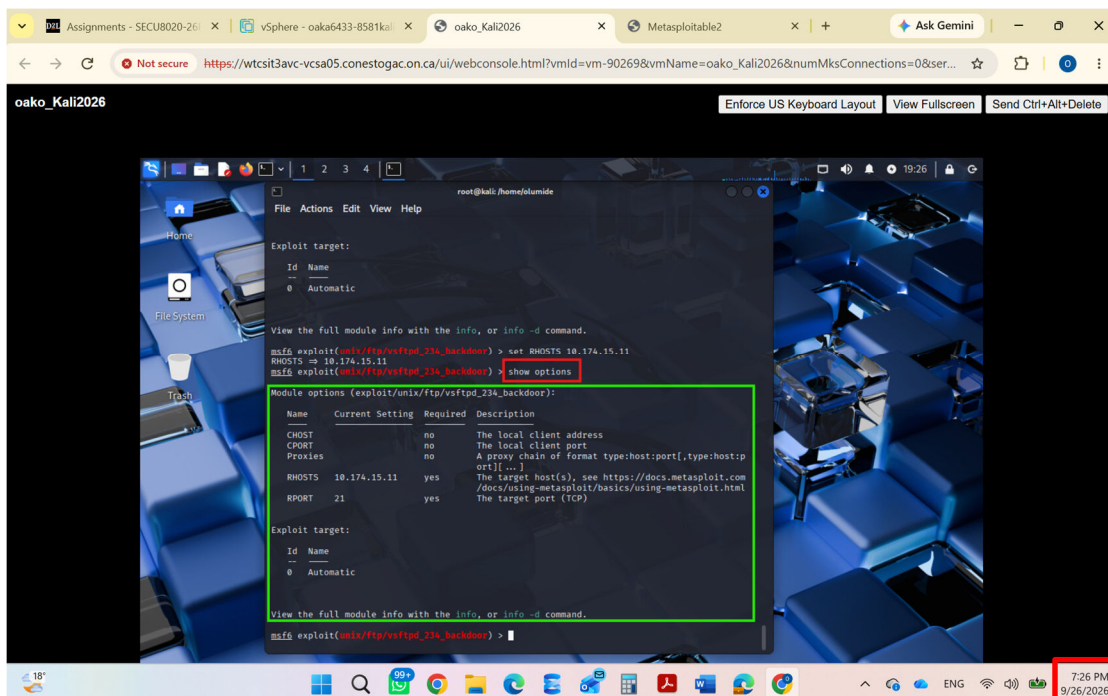


Image 12: Confirming the right options was set with the *show options* command, output shows that Remote host 10.174.15.11 has been set

Task 7

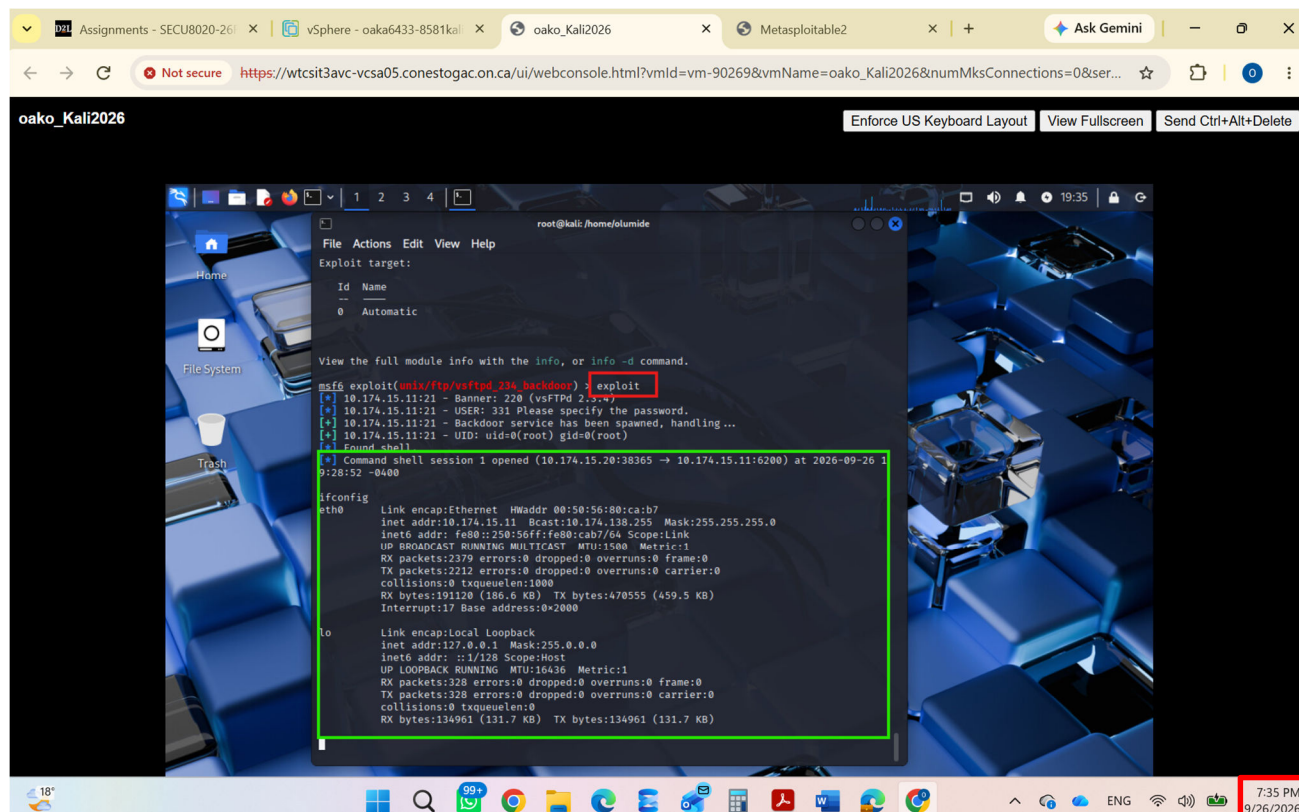


Image 13: The *exploit* command is used to take over the Metasploitable2 VM and running *ifconfig* command shows that I now have access to it.

Task 8

Reflection

What I learned from this exercise was a lot more about how a real exploitation process operates in practice, starting with network discovery, identifying a vulnerable service, identifying the version of the vulnerable service, finding an appropriate Metasploit module, configuring the module to fit the needs of the process, and executing the module, thereby taking control of the system. It is one thing to read about the vsftpd 2.3.4 backdoor vulnerability, but performing search, use, show options, set, and exploit in Metasploit showed the practicalities of the attack, and the threat that comes from running a vulnerable, unpatched service in business.